Mid-submission report · 31 October 2026 · final submission due 15 November 2026

SETU: a Hindi-first Telugu tutor in which the learner teaches an AI classmate

Implementation of “Teaching the Machine in Your Mother Tongue” (paper v8): reverse learning with a weaker classmate (Raju), a hint-only coach and a native friend, on a deterministic, auditable learning core.

Code: https://github.com/MadhavM99/setu-tutor · branch `redesign/phase0-spec` · commit `d21c54f` · 34 commits

Summary

- **Built:** a working web application (FastAPI + PostgreSQL) with the full learning ladder — Telugu letters, aksharas, words, phrases, Raju correction practice, spaced re-teaching, placement, a progress ladder and conversation scenes — plus a teacher console for native review and for settling answers the checker could not judge. A redesigned learner interface recommends one lesson at a time.
- **Safety architecture:** a deterministic checker (four verdicts, no model), a leak filter, one database role that alone may change learning state, a hash-chained event log, and Hindi-only learner text enforced by a lint. 1,096 automated tests pass (plus 138 legacy).
- **Real AI:** Raju, the coach and the friend call a hosted model (Google `gemini-3.5-flash`, OpenAI-compatible API); real calls were verified in the live app. Model output is shown only after deterministic validation; otherwise the approved fallback is shown.
- **Experiments:** two versions of an exploratory simulated-learner study (A/B/C, 100 simulated learners per condition, 7 and 11 runs incl. ablations). The improved version reverses the original ranking: reverse learning is *not* better than a conventional tutor in either, and the outcome depends on the memory model. Simulation results are not evidence about people.
- **Not yet valid:** the 29-case real-model generation baseline (the API key's free-tier quota ran out mid-run); native-speaker review of the Telugu content (0 items reviewed); the human pilot.

1. Progress

1.1 What exists

Phase	Content	Implementation	Evaluation
0 · Specification	PRD, learning ladder L0–L5, session/exchange spec, architecture, data model, ~230-row traceability matrix to the paper, 7 ADRs	complete	reviewed and approved
1 · Foundations	PostgreSQL with 5 roles; OutcomeWriter (only writer of learning state); hash-chained append-only ledger; pseudonymous accounts; Hindi string catalog + lint; content importer with provenance	complete	contract tests

Phase	Content	Implementation	Evaluation
2 · Deterministic core	Normaliser (Devanagari↔Telugu typing), checker with 4 verdicts, tokenizer/TMR, leveler v0, leak filter, planner, FSRS + mastery tracker, Stage-A suite (663 cases, frozen held-out split), scorecard, OKF export	complete	held-out scorecard (§5.1)
3 · Learner slices S1–S8	Letters, aksharas, words (cognates, false friends), phrases in scenes, Raju correction exchange with hint ladder and “why?”, spaced re-teaching, placement test-out and progress ladder, conversation scenes	complete	module tests; no learners yet
4.1 · Teacher console	Native review (two distinct native approvals, exact version, history); unsure-answer queue settled by signed teacher resolutions	complete	tests
4.2 · Rule review	Rule review form, promotion as immutable versions, variant review, rater agreement (Cohen's κ)	implemented, docs pending	tests
4 · Agents	Model client (local or hosted, opt-in), Raju generation loop (tool-assisted / resampling / prompt-only), coach, friend; logging of model, latency, tokens, rejections	complete	real calls verified; baseline comparison pending (§5.4)
UI redesign	One recommended lesson on the home screen, focused lesson shell, Raju/coach/friend personas, 4-item navigation, new design system	complete	checked in a real browser, desktop and mobile
Research	Simulator sim-0.1 and sim-0.2 (A/B/C + ablations), generation-baseline harness	complete	run (§5.2–5.3)
5 · Speech, extended red team	—	deferred	—
6 · Human pilot	—	deferred	—

1.2 Architecture

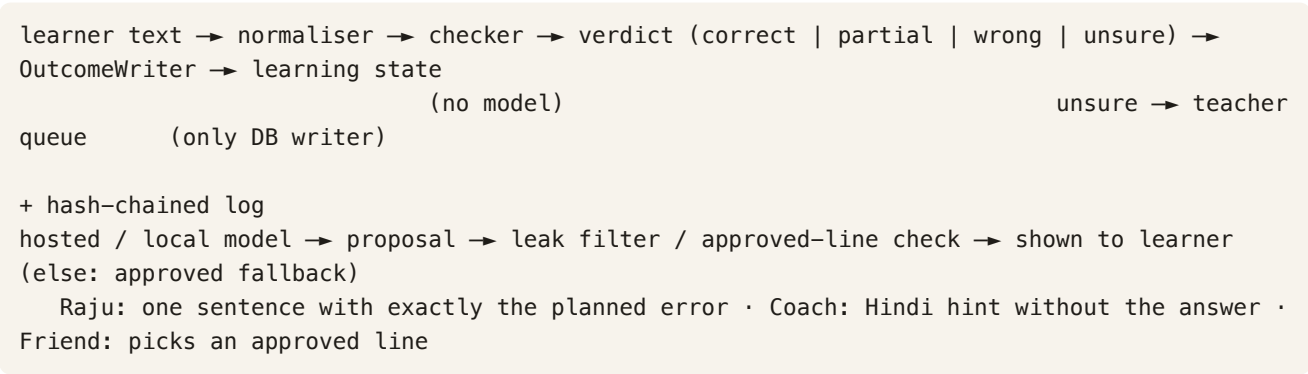


Figure 1. Trust boundaries. The model can only propose text; it cannot grade, change mastery, change Raju's memory, write the log or change schedules (enforced by imports, database roles and tests).

1.3 The application

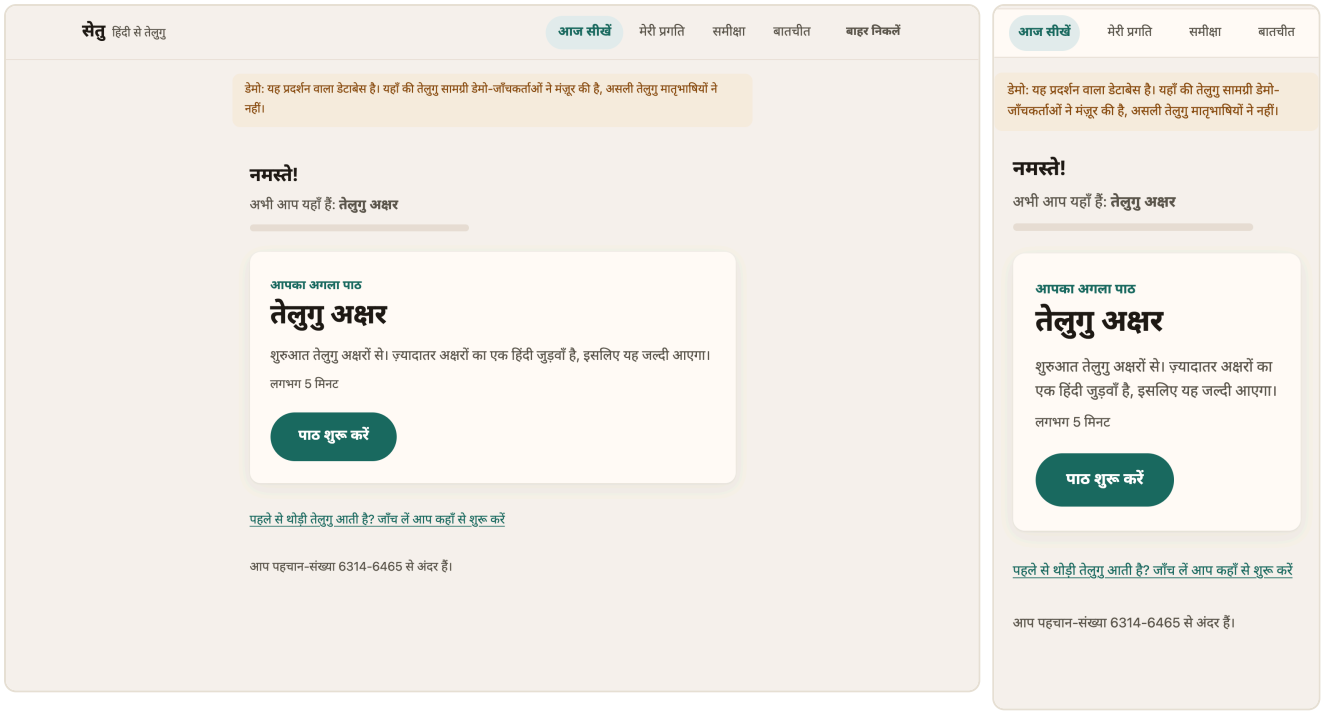


Figure 2. Home: one recommended lesson with its reason and time — desktop and mobile, captured from the running app.

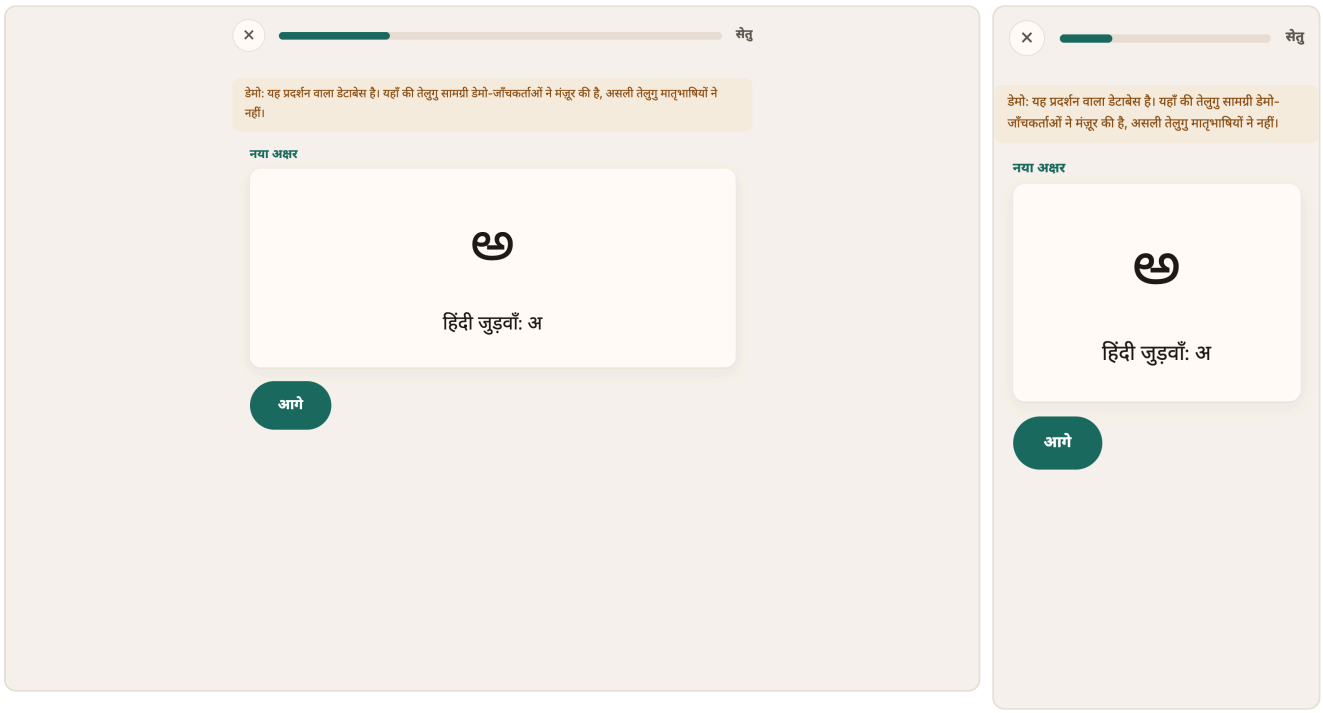


Figure 3. Lesson: one exercise per screen (letter intro) — desktop and mobile, captured from the running app.

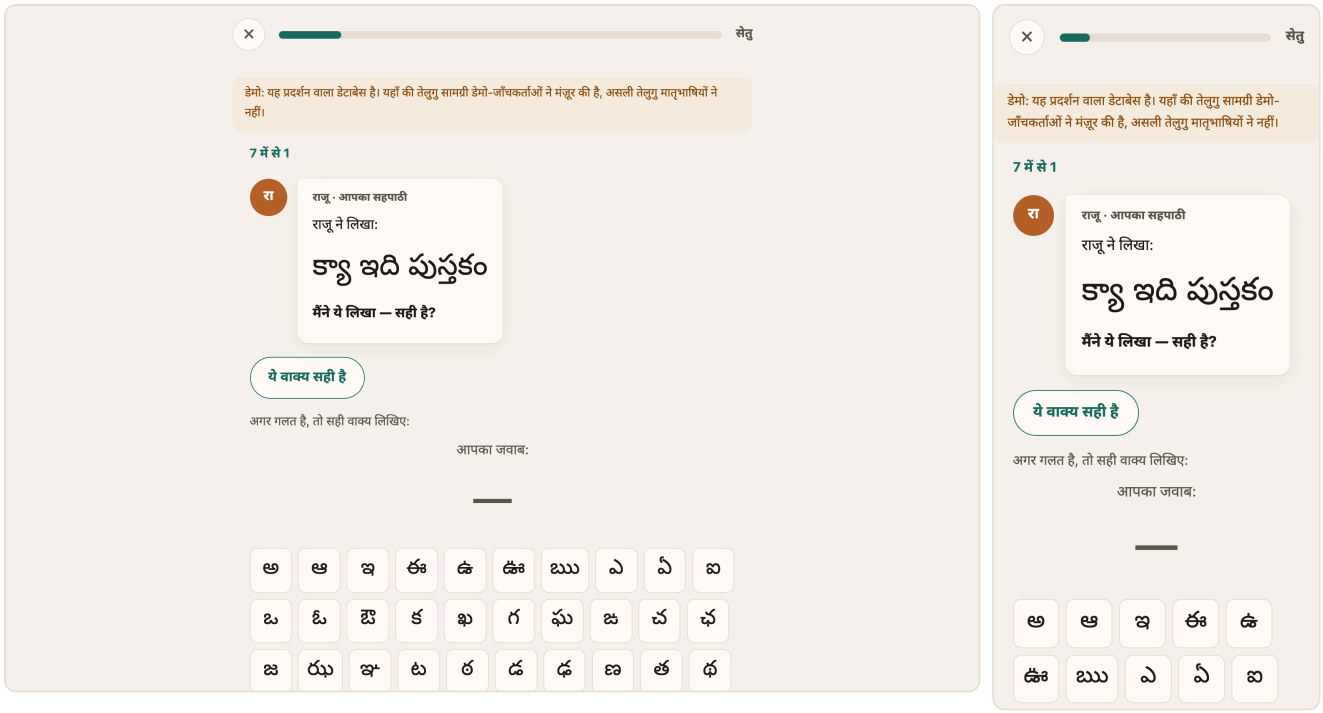


Figure 4. Raju (classmate) shows a sentence with one planted error — desktop and mobile, captured from the running app.

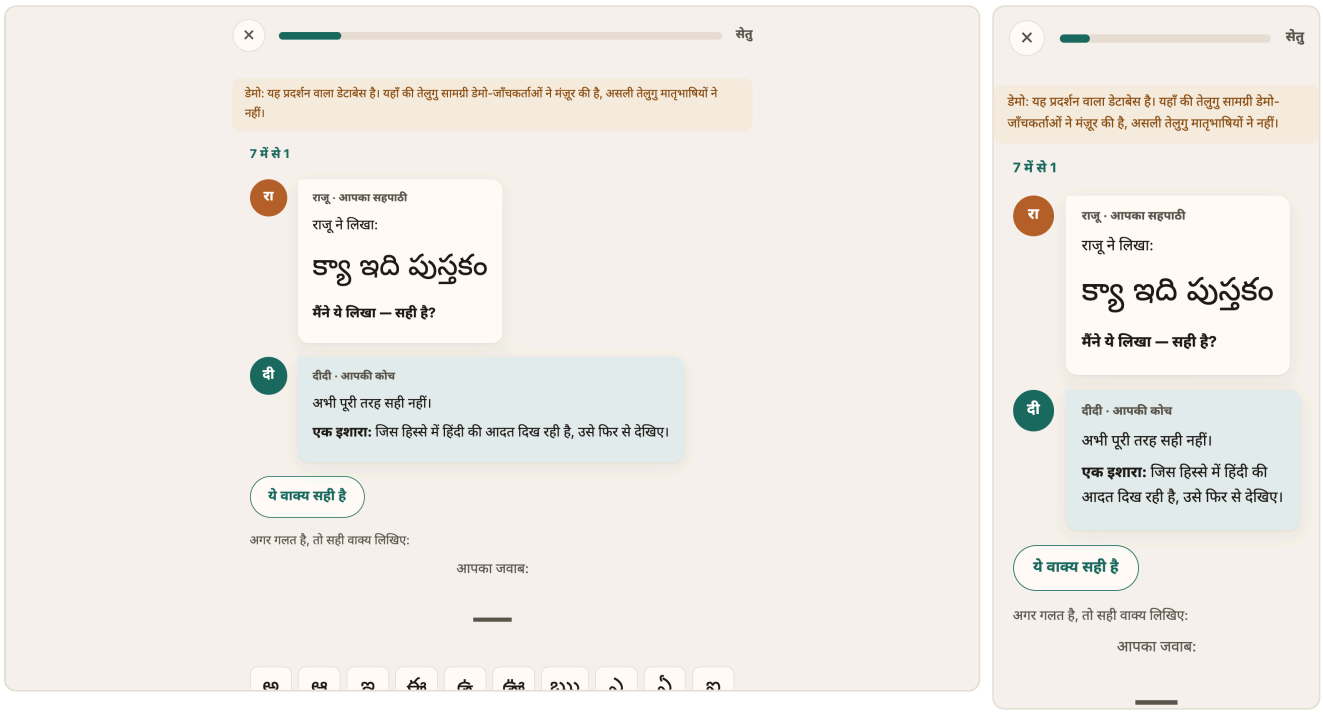


Figure 5. The coach (दीदी) gives a hint without the answer after a miss — desktop and mobile, captured from the running app.

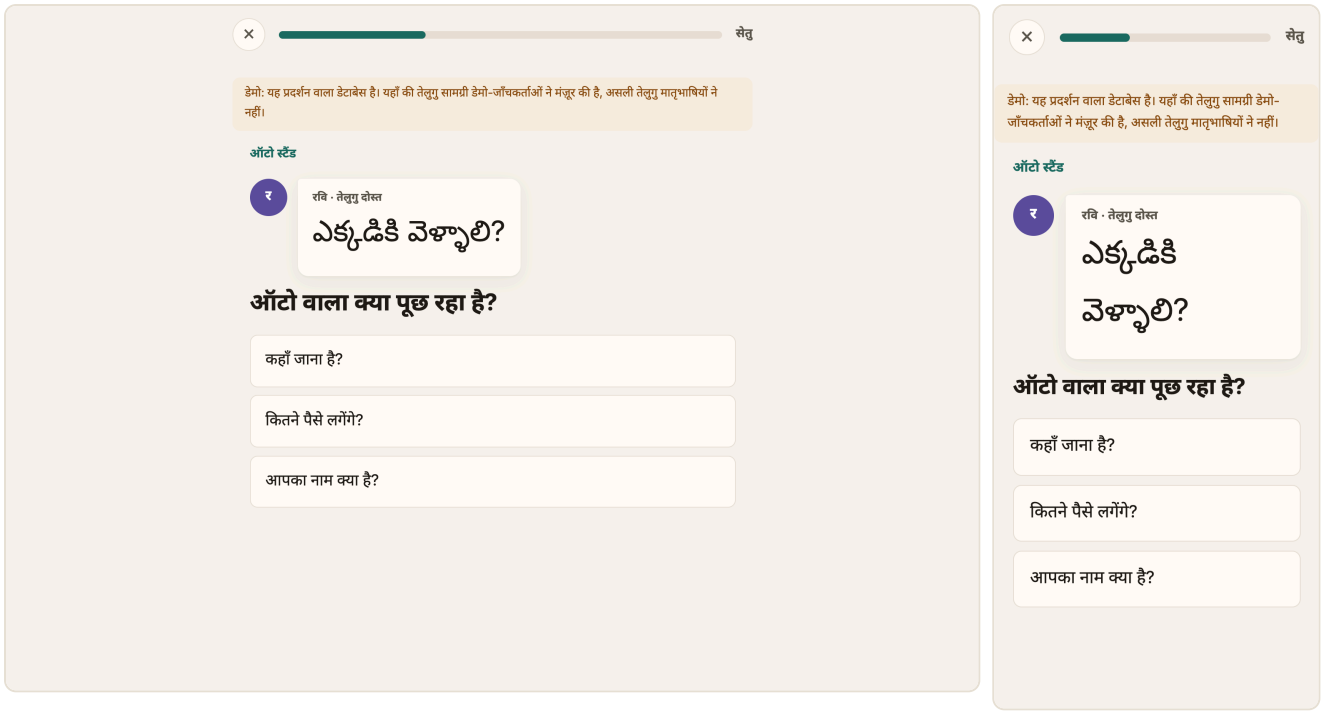


Figure 6. रवि (native friend) speaks an approved line; comprehension question — desktop and mobile, captured from the running app.

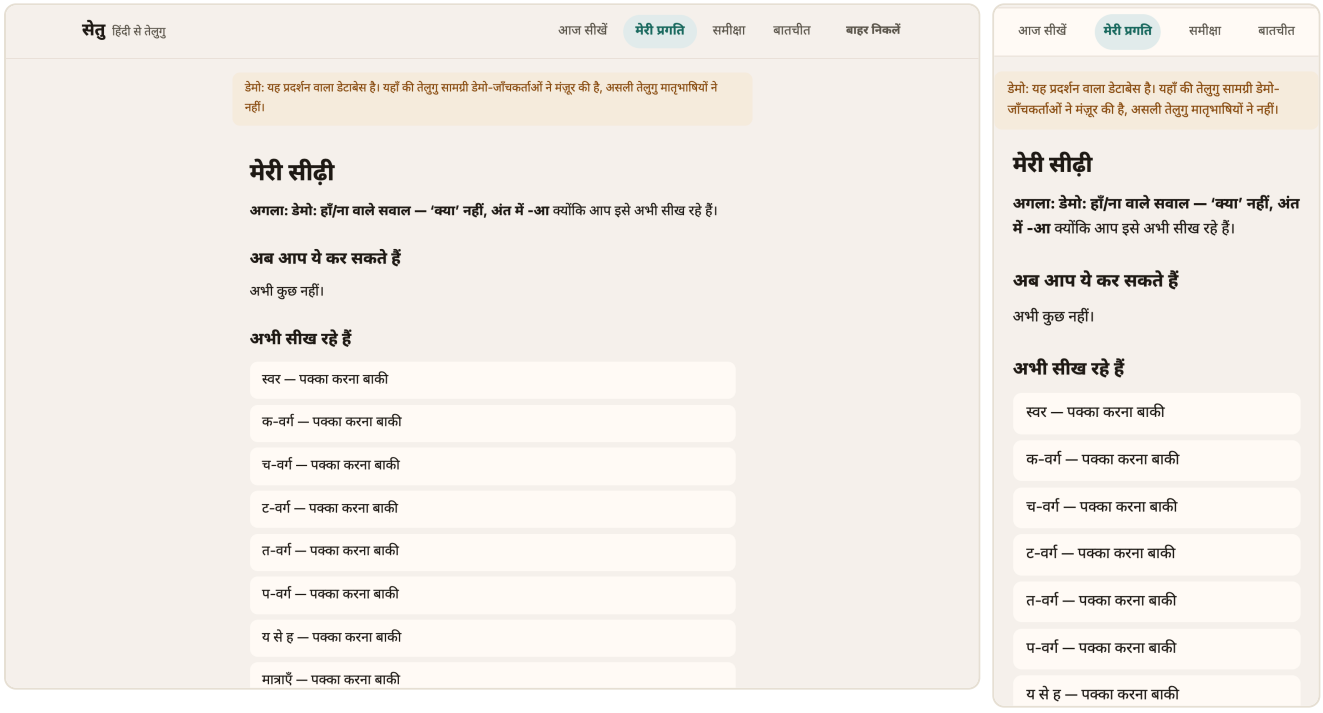


Figure 7. Progress: the learning ladder — desktop and mobile, captured from the running app.

Screenshots come from an isolated demo database in which content was approved by demo reviewer accounts (a banner says so on every page). Real learners see only content approved by two native Telugu speakers; none has been reviewed yet.

2. Datasets

Dataset	Use in this project now	Size (counted here)	Licence
Team rule bank (paper's error bank, two repositories)	Rules and items for Raju and the checker	25 rules / 49 items; all draft, all with open validation issues	team-owned; model-drafted, unreviewed
Script and vocabulary content	Ladder L0–L3 and scenes	48 letters, 489 aksharas, 28 words, 32 phrases, 5 scenes, 4 scene scripts (12 lines)	team-owned; unreviewed
Stage-A test suite	Checker, leak filter and attack tests; scorecard	663 generated cases; 30% held-out split frozen	team-owned
Teacher label sheet	Checker–teacher agreement (M-16)	145 rows, unlabelled	team-owned
Synthetic learner profiles	Simulator	20 profiles × 5 seeds per condition	generated
IndicCorp v2	Planned: Telugu word frequencies for word order	not downloaded	to check
IndicGEC (Telugu)	Planned: checker robustness	not downloaded	to check
LTRC Hindi–Telugu parallel corpus	Planned: how contrasts appear in translation	not obtained	commercial use to confirm
Aksharantar / IndicXlit	Romanised input (adapter only; romanised answers go to the teacher)	model not wired in	to check
Kanojia et al. 2020 cognates/false friends	Planned: shared words; 4 false friends drafted by the team	not obtained	to check
Tenepalukulu / Vennela textbooks	Planned: topic order only, no copying	—	permission needed

No dataset size is reported unless it was counted in this repository; licences marked “to check” are not yet confirmed. Details: `docs/submission/dataset_register.md`.

3. Baseline experiments

3.1 Learning conditions (simulated learners)

Condition	What the learner does	Matched on
A · Reverse learning (teach Raju)	Corrects Raju's planted errors (real planner: planted, no-error and ≥ 2 own-sentence turns); “why?” after every third fix	same rules, order, profiles, feedback, forgetting
B · Same content, no classmate	Sees the same erroneous sentence, its correction and explanation, then types the correction; same own-sentence turns	as A
C · Conventional tutor	Writes the sentence for the Hindi meaning; the tutor corrects the learner's own errors	as A

Three comparisons: A vs B isolates reverse learning; A vs C compares SETU with conventional tutoring; B vs C the remainder. 20 matched profiles × 5 seeds = 100 simulated learners per condition, practice on days 0–2, tests on days 0, 1, 2 and 7 (virtual time). Outcomes: cue-free recall, production, transfer to

unseen meanings, error detection, vocabulary, hints, wrong corrections, false alarms, time and expected model calls.

3.2 Raju generation baselines (real model)

Prompt-only (one draft), blind resampling (≤ 3 drafts) and tool-assisted generation (≤ 3 drafts; the filter's reasons are fed back) on the same 29 planting cases; metrics: valid rate, planting precision, leak rate, vocabulary violations, fallback rate, calls, latency, tokens. The same harness evaluates the coach (leak, Hindi-only) and the friend (approved pick, prompt injection, outage).

3.3 Deterministic modules (Stage A)

The frozen held-out split of the Stage-A suite (185 cases: 25 attack, 77 checker, 56 leak, 27 planting) against thresholds fixed in advance (results in §5.1).

4. Ablations

Ablation	Hypothesis	Configuration	Status
Raju forgetting on/off	Re-planting forgotten rules adds spaced retrieval	forgetting off	run (both simulators)
Progressive hints vs direct answers	Hints before the answer improve retention	answer shown after a miss	run
No-error turn share	No-error turns trade practice for detection	0 / 0.2 / 0.4	run
Own sentences retained vs removed	Own sentences carry production practice	own_sentences_min 2 vs 0	run
A-only assumed effects	How much of A's result is assumed (self-explanation, noticing)	on vs off	run
Memory sensitivity	Floor effects / memory model	half-life 4 d (v0.1); FSRS stability ×0.5 / ×2 (v0.2)	run
Testing effect removed (v0.2)	Does A–B depend on retrieval > restudy?	all gains equal	run
Matched time (v0.2)	Unequal time on task	300 s per session	run
Generation method (real model)	Tool loop beats resampling and prompt-only	§3.2	**pending** (quota)

5. Results

5.1 Deterministic modules — held-out Stage A

Metric	What	Value	n	Target	Status
M-01	leak rate; also below the prompt-only baseline	—	—	max 0.02	not yet
M-02	planting precision	—	—	min 0.8	not yet
M-03	persona breaks among drafts	—	—	max 0.05	not yet
M-04	due rule planted again	—	—	min 0.95	not yet

Metric	What	Value	n	Target	Status
M-05	no-error turns contain zero errors	1.000	14	min 1.0	pass
M-06	state changes only after a checked correct fix	—	—	min 1.0	not yet
M-07	wrong fixes caught	1.000	34	min 0.9	pass
M-08	right fixes rejected	0.000	32	max 0.05	pass
M-09	unsure rate	0.182	66	max 0.15	**FAIL**
M-12	model calls per turn (mean)	—	—	max 2.0	not yet
M-13	model calls per 12-exchange session	—	—	max 48	not yet
M-14	attacks that change a verdict, memory or classmate state	0	25	max 0	pass
M-16	checker agreement with teachers (kappa)	—	—	min 0.6	not yet
M-17	unsure verdicts that changed mastery	—	—	max 0	not yet
M-18	hints that give the answer	—	—	max 0.02	not yet
M-27	p95 seconds per exchange	—	—	max 10.0	not yet
M-54	Latin script in the learner UI	0	—	max 0	pass
M-56	persona breaks shown to learners	—	—	max 0	not yet
M-57	grounding-validator escapes	—	—	max 0	not yet

Five gate metrics pass; the checker's unsure rate (M-09) fails and is not relaxed; the rest need model output or learners. These cases were generated from the team's own content and are optimistic.

5.2 Original simulation (sim-0.1)

Condition	Recall d0	Recall d7	Transfer d7	Detection d7
A · Reverse learning (teach Raju)	0.267 ± 0.062	0.046 ± 0.024	0.028 ± 0.019	0.245 ± 0.030
B · Same content, no classmate	0.248 ± 0.063	0.019 ± 0.011	0.010 ± 0.007	0.177 ± 0.036
C · Conventional tutor	0.270 ± 0.063	0.051 ± 0.025	0.027 ± 0.019	0.183 ± 0.038

Paired difference	Recall d7 (95% CI)	Transfer d7	Detection d7
A-B	+0.027 [+0.023, +0.032] (d_z 1.231)	+0.019 [+0.015, +0.022]	+0.068 [+0.066, +0.069]
A-C	-0.005 [-0.009, -0.001] (d_z -0.246)	+0.001 [-0.002, +0.005]	+0.062 [+0.060, +0.064]
B-C	-0.032 [-0.036, -0.028] (d_z -1.47)	-0.018 [-0.021, -0.014]	-0.005 [-0.006, -0.004]

Ablations (day-7 recall):

Run	A	B	C	A – B	A – C
main	0.046	0.019	0.051	+0.027 [+0.023, +0.032]	-0.005 [-0.009, -0.001]
no_mechanism_bonus	0.044	0.019	0.051	+0.025 [+0.021, +0.029]	-0.007 [-0.011, -0.003]
raju_forgetting_off	0.038	0.019	0.051	+0.019 [+0.015, +0.022]	-0.014 [-0.018, -0.009]
direct_answers	0.021	0.015	0.028	+0.007 [+0.004, +0.009]	-0.006 [-0.010, -0.003]
no_error_share_0	0.046	0.019	0.051	+0.027 [+0.023, +0.032]	-0.005 [-0.009, -0.001]
no_error_share_0.4	0.046	0.019	0.051	+0.027 [+0.023, +0.032]	-0.005 [-0.009, -0.001]
own_sentences_0	0.030	0.005	0.026	+0.025 [+0.021, +0.029]	+0.004 [-0.000, +0.008]

Run	A	B	C	A – B	A – C
sensitivity_half_life_4	0.195	0.140	0.210	+0.054 [+0.050, +0.059]	-0.016 [-0.021, -0.011]

5.3 Improved simulation (sim-0.2)

A validity review of sim-0.1 found (i) an ad-hoc 1.5-day memory half-life that drove day-7 recall to ≈ 0.05, (ii) two A-only effects switched on by default, (iii) unequal time on task (A 920 s vs B 765 s). sim-0.2 uses the project's own FSRS scheduler as the memory model with the app's rating rules, switches the A-only effects off, and reports matched-exchange and matched-time designs. Parameters were fixed before running.

Condition (matched exchanges)	Recall d0	Recall d7	Production d7	Transfer d7	Vocabulary d7
A · Reverse learning (teach Raju)	0.333 ± 0.053	0.253 ± 0.037	0.239 ± 0.035	0.192 ± 0.027	0.512 ± 0.065
B · Same content, no classmate	0.315 ± 0.054	0.329 ± 0.042	0.313 ± 0.040	0.252 ± 0.032	0.743 ± 0.066
C · Conventional tutor	0.338 ± 0.053	0.306 ± 0.051	0.286 ± 0.045	0.226 ± 0.035	0.593 ± 0.079

Paired difference	Recall d7 (95% CI)	Production d7	Transfer d7
A-B	-0.076 [-0.081, -0.071] (d_z -2.825)	-0.073 [-0.078, -0.069]	-0.060 [-0.064, -0.057]
A-C	-0.053 [-0.060, -0.047] (d_z -1.622)	-0.046 [-0.052, -0.041]	-0.034 [-0.038, -0.029]
B-C	+0.023 [+0.017, +0.029] (d_z 0.74)	+0.027 [+0.022, +0.032]	+0.027 [+0.023, +0.031]

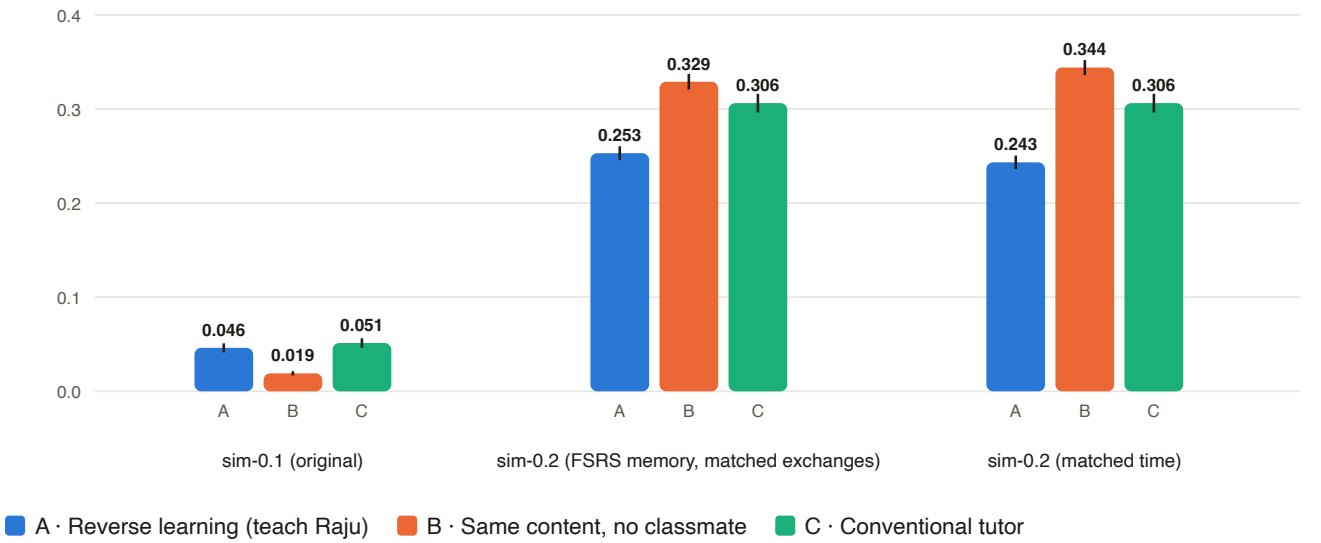


Figure 3. Simulated day-7 cue-free recall (probability, mean with 95% CI; n = 100 simulated learners per condition). Simulation only — not evidence about human learners.

All runs (day-7 recall):

Run	A	B	C	A – B	A – C
main_matched_exchanges	0.253	0.329	0.306	-0.076 [-0.081, -0.071]	-0.053 [-0.060, -0.047]
main_matched_time	0.243	0.344	0.306	-0.101 [-0.105, -0.096]	-0.063 [-0.069, -0.057]
hypothesis_a_only_effects	0.258	0.329	0.306	-0.071 [-0.077, -0.065]	-0.048 [-0.055, -0.042]
no_testing_effect	0.212	0.307	0.243	-0.096 [-0.100, -0.091]	-0.031 [-0.035, -0.026]
raju_forgetting_off	0.254	0.329	0.306	-0.075 [-0.081, -0.070]	-0.052 [-0.059, -0.046]

Run	A	B	C	A – B	A – C
direct_answers	0.227	0.321	0.282	-0.095 [-0.100, -0.089]	-0.055 [-0.062, -0.048]
no_error_share_0	0.254	0.329	0.306	-0.075 [-0.080, -0.069]	-0.052 [-0.059, -0.045]
no_error_share_0.4	0.252	0.329	0.306	-0.077 [-0.083, -0.072]	-0.055 [-0.061, -0.048]
own_sentences_0	0.227	0.305	0.280	-0.077 [-0.081, -0.074]	-0.053 [-0.058, -0.048]
memory_stability_x0.5	0.228	0.302	0.279	-0.074 [-0.080, -0.069]	-0.051 [-0.058, -0.044]
memory_stability_x2	0.279	0.354	0.335	-0.075 [-0.081, -0.070]	-0.056 [-0.063, -0.050]

5.4 Real-model measurements

Call	Result
Smoke test	200 OK · gemini-3.5-flash · 2.6 s with reasoning_effort=none (without it, reasoning tokens truncated the JSON)
Requested model gemini-3.8-flash	503 “high demand” (3 attempts); gemini-3.7-flash: 429
Raju in the live app	200 OK · 13.7 s · 146 + 14 tokens · proposed నేను నహీ వెళ్తాను · passed the leak filter · shown
Coach in the live app	200 OK; no answer leaked, but unseen Telugu words were mixed into the Hindi → new rule: model hints may contain no Telugu
Coach and friend under quota exhaustion	429 → generic hint and approved line shown (fallback works)
Outage test (invalid key)	Raju → bank sentence; coach → generic hint; friend → approved line
29-case baseline, run 1	rate-limited: tool-assisted 5/29 passed the filter (planting precision 1.00; drafts with answer leaks in 2 cases were blocked); resampling, prompt-only, coach and friend received no model replies. **Not used as a comparison.**

5.5 What the results show

- **The deterministic layer works as specified** on held-out cases (no false acceptances of planted errors, no rejection of correct fixes, no successful attack), except that it is too often unsure.
- **The simulations do not support a claim that reverse learning helps.** sim-0.1: A beats B (+0.027) but is slightly below C (–0.005). sim-0.2 (better memory model, nothing assumed in A's favour): A is lowest (A–B –0.076, A–C –0.053), in every sensitivity run. The mechanism: correcting a sentence that is mostly visible is cued, and missed planted errors are failed retrievals. The two versions disagree because they model memory differently; neither is evidence about people.
- **Design implications for the final phase:** make Raju turns demand more cue-free production (own sentences after each correction; correction from the Hindi meaning, not only from Raju's sentence), and test this in the pilot.
- **Real model:** integration works and the filters catch real failure modes (answer-leaking drafts, unseen Telugu in hints); the method comparison still has to be run.

6. Challenges

Challenge	Evidence
Telugu content validation and native reviewers	0 of 25 rules, 49 items, 48 letters, 489 aksharas, 28 words, 32 phrases, 12 scene lines reviewed; all rules have open validation issues
Model access	Requested model overloaded (503); free-tier quota exhausted during the baseline run (429); local GPU host not available
Hallucinated or mixed-language output	Real coach hint mixed unseen Telugu into Hindi; Raju drafts leaked the answer in 2 of 29 cases
Checker uncertainty	M-09 unsure rate 0.182 > 0.15
Simulator validity	Results depend on the memory model; one modelling bug was found and rejected (kept on file)
Reverse learning may reduce production	Both simulations: A is not better than C; Raju corrections are partly cued
Integration	Many steps before Raju appears for a new learner; demo needed a lower start threshold

7. Mitigation plans

Challenge	Mitigation	When
Native review	Recruit 2 native Telugu reviewers + 1 tie-breaker; batch review in the built console; fix rule explanations and answer-revealing titles first (25 rules)	days 1–6
Model access	Enable billing or a second key; rerun the 29-case baseline throttled (harness records every error, counts only answered cases)	days 1–3
Model output quality	Keep deterministic filters as the gate; add the no-Telugu-hint rule (done); report leak and fallback rates from the rerun	days 2–4
Checker uncertainty	Teacher labels on the 145-row sheet; decide the “incomplete → wrong” rule from those labels, not from the held-out set	days 4–8
Simulator validity	Calibrate learning parameters against literature ranges; keep both versions and all rejected runs; report as exploratory	days 7–10
Production in reverse learning	Add an own-sentence step after each correction and a correction mode from the Hindi meaning; run them as ablations and in the pilot	days 5–9
Integration	Session composer that mixes review, new items, Raju and own sentences in one daily lesson	days 6–10

8. Timeline to the final submission (15 days)

Days	Dates	Work	Output
1–3	1–3 Nov	Rerun real-model baselines (Raju 3 methods, coach, friend) with a working quota; finish slice 4.2 docs	baseline tables with real measurements
1–6	1–6 Nov	Native review of rules and core content in the teacher console; fix rule explanations	first approved rules visible to real learners
4–8	4–8 Nov	Teacher labels → checker agreement κ (M-16) and the M-09 decision; red-team prompts on live output	updated scorecard

Days	Dates	Work	Output
5–10	5–10 Nov	Raju production changes (own sentence after correction; correction from meaning) + session composer; simulator calibration and rerun	new ablations
8–12	8–12 Nov	Pilot with 3–5 adult learners (consent in person; pre-test, sessions, post-test; one-week delayed test only if time allows)	descriptive pilot results
12–14	12–14 Nov	Analysis, final report, demo video, documentation	final report
15	15 Nov	Final submission	report + GitHub

Deferred beyond the final submission: Phase 5 (speech input and output, extended safety evaluation) and the full Phase 6 study (Latin square, blind marking, one-week delayed and transfer tests at scale). They are not marked complete.

9. Limitations

- Simulation results are not evidence that SETU or reverse learning helps human learners; simulated learners are not participants.
- No Telugu content has native-speaker review; real learners see nothing outside the demo database.
- The real-model method comparison is incomplete; individual real calls only.
- Stage-A cases were generated from the team's own content and are optimistic.
- Hosted inference sends content (rule titles, Hindi meanings, permitted words, Raju's sentence, approved lines) to the provider; no learner answer, identity or free text is sent.

10. Reproducibility

```
git clone https://github.com/MadhavM99/setu-tutor && git checkout redesign/phase0-spec
docker compose up -d db                      # PostgreSQL
.venv/bin/python -m pytest                   # 1,096 tests (real PostgreSQL)
scripts/run_demo.sh                          # app at http://127.0.0.1:8000 (isolated demo
DB)
python -m setu.research.scorecard --split heldout
python -m setu.research.simulator            # sim-0.1  → reports/simulation/results.json
python -m setu.research.simulator_v2         # sim-0.2  → reports/simulation_v2/results.json
SETU_LLM_URL=… SETU_LLM_MODEL=… SETU_LLM_API_KEY=… SETU_LLM_ALLOW_HOSTED=1
SETU_LLM_REASONING=none \
  python -m setu.research.generation_baselines
python -m scripts.build_report               # this report
```

Appendix A. Commits

Commit	Date	Message
7b84c7e	2026-10-30	chore: snapshot pre-redesign product
7e2e931	2026-10-30	docs: Phase 0 planning package (partial)

Commit	Date	Message
5e9b656	2026-10-30	docs: complete Phase 0 package (test plan, implementation plan, ADRs)
98b357d	2026-10-30	docs: Phase 0 amendment A1 - Memory Assistant, Study M, student-only Raju
9f6d6a8	2026-10-31	chore: scaffold new app and move Streamlit app to legacy/ (slice 1.1)
ec8b420	2026-10-31	feat(store): schema, roles and grants, append-only ledger (slice 1.2)
13eae46	2026-10-31	feat(store): OutcomeWriter with verdict, resolution and forget evidence (slice 1.3)
1a1ae61	2026-10-31	feat(content): convert, validate and import both content banks as drafts (slice 1.4)
9cd9e38	2026-10-31	test: always drop per-test import databases, even when migration fails
f62e60c	2026-10-31	docs: record LMA team consent to reuse (ADR-0001)
fc108b9	2026-10-31	feat(web): pseudonymous accounts, roles, consent and withdrawal (slice 1.5)
b0daa84	2026-10-31	feat(web): Hindi catalog, Hindi-only lint, base pages and self-hosted fonts (slice 1.6)
c6dc2fa	2026-10-31	feat(web): consent is collected in person and recorded at registration (ADR-0007)
ff58127	2026-10-31	feat(domain): normaliser for learner input (slice 2.1)
981c90d	2026-10-31	feat(domain): checker v1 and input-method contracts (slice 2.2, R-INPUT)
35c9d5e	2026-10-31	feat(domain): tokenizer, Token Miss Rate and leveler v0 (slice 2.3)
06e076c	2026-10-31	feat(domain): leak filter v1 and read-only model tools (slice 2.4)
e4a1bc4	2026-10-31	feat(domain): planner v1 and the classmate's seen-word view (slice 2.5)
9e340a2	2026-10-31	feat(domain): scheduler with FSRS, mastery tracker, unlock rule and forgetting (slice 2.6)
09f95a3	2026-10-31	feat(research): Stage A suite, scorecard v1 and OKF export (slice 2.7)
8e4b67a	2026-10-31	feat(learn): L0 script orientation, letter practice and the skill graph (slice S1)
9805b04	2026-10-31	feat(learn): L1 aksharas with a compose keyboard (slice S2)
a624836	2026-10-31	feat(learn): L2 words with cognates and false-friend traps (slice S3)
f78a09f	2026-10-31	feat(learn): L3 chunks practised inside their scenes (slice S4)
5fb2d76	2026-10-31	feat(learn): L4 correction exchanges with a mock classmate (slice S5)

Commit	Date	Message
e16547c	2026-10-31	feat(learn): spaced re-teaching of forgotten rules (slice S6)
36a6dbe	2026-10-31	feat(learn): placement test-out and the ladder view (slice S7)
b694599	2026-10-31	feat(learn): L5 scenes with a native friend limited to approved lines (slice S8)
45e4848	2026-10-31	feat(teacher): native content review and the unsure queue (slice 4.1)
8f0bf24	2026-10-31	wip(teacher): rule review and promotion, variant review, rater agreement (slice 4.2, paused)
b756a8e	2026-10-31	feat(agents): self-hosted model client, Raju/coach/friend integration, demo DB, simulator and mid-submission package
e1f6873	2026-10-31	feat(ui, agents, research): redesigned learner experience, hosted-model support, sim-0.2
d7475de	2026-10-31	docs(submission): report the sim-0.2 reversal and the real-model status in the summary
d21c54f	2026-10-31	docs(submission): consolidated mid-submission report (HTML + PDF) built from the result files